Fiscal Year:	FY 2012	**Task Last Updated:**	FY 07/09/2012
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Project Title:	Modifying Protein-Lipid Interactions of a Microbial Mechanosensitive Channel: Functional Screens after Tethering Specific Residues to the Membrane		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
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Task Description:	At the foundation of the ability of organisms to detect gravity is the capacity to detect mechanical forces. It is, for humans, the sheering force on stereocilia of vestibular hair cells, and for plants, the sedimentation of amyloplasts within a root cell. One of the primary molecules underlying the ability to detect mechanical forces is the mechanosensitive channel. To date, the best characterized is MscL, a microbial MechanoSensitive Channel of Large conductance. MscL opens a large pore in response to membrane tension; several studies now suggest sensing membrane tension is common among many mechanosensitive channels from microbes to plants and animals. Hence, the interactions between mechanosensitive channels and the lipid-membrane environment are of tremendous interest. Manipulating these interactions should have functional consequences that yield important clues of how the channel normally interacts with lipid. We have generated and characterized a library of mutated MscL channels in which each residue in and around the transmembrane domains has been sequentially mutated to cysteine. Treatment of these mutants with sulfhydryl reagents containing aromatic rings or alkyl chains should encourage hydrophobic interactions or tether the residue to the membrane. If the modified residue is normally within a hydrophobic environment primarily when in a specific state (closed, transition or open), then that state would be stabilized. We have developed an in vivo screen that allows us to rapidly assess functional changes in the MscL channel simply by monitoring bacterial growth; channels in which the open, closed, or even various transition states have been stabilized are identified. Once manipulations that lead to functional consequences are selected, the exact functional changes of MscL will be assessed by electrophysiological assays. Preliminary experiments have demonstrated the feasibility of this approach. When completed, this study will yield critical insight into how mechanosensitive channels sense and respond to membrane tension.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many mechanosensors, most likely including those involved with gravity sensing, appear to sense and respond to biophysical properties of the lipid environment. Thus, we have been studying the protein-lipid interactions of a bacterial mechanosensor by using an in vivo screen (as described in detail in the proposal). We have found a “hot spot” near the second transmembrane domain, just proximal to a cytoplasmic cluster of charged amino acids. The data suggest that the residues just proximal to this cluster enter the lipid environment upon sensor stimulation. This finding may have implications for the functioning of other sensors: a cytoplasmic charge cluster is a common feature among many channels, including candidate mechanosensors. We speculate that the movement of this region into the membrane could be an underlying feature for the function of many mechanosensors. We recently had a manuscript accepted as a minireview for the Biophysical Journal that emphasizes these points. It is entitled "Sensing and responding to membrane tension: the bacterial MscL channel as a model system."
Task Progress:	Summaries of Research, and Education Activity Report: Original summary: At the foundation of the ability of organisms to detect gravity is the capacity to detect mechanical forces. It is, for humans, the sheering force on stereocilia of vestibular hair cells, and for plants, the sedimentation of amyloplasts within a root cell. One of the primary molecules underlying the ability to detect mechanical forces is the mechanosensitive channel. To date, the best characterized is MscL, a microbial MechanoSensitive Channel of Large conductance. MscL opens a large pore in response to membrane tension; several studies now suggest sensing membrane tension is common among many mechanosensitive channels from microbes to plants and animals. Hence, the interactions between mechanosensitive channels and the lipid-membrane environment are of tremendous interest. Manipulating these interactions should have functional consequences that yield important clues of how the channel normally interacts with lipid. We have generated and characterized a library of mutated MscL channels in which each residue in and around the transmembrane domains has been sequentially mutated to cysteine. Treatment of these mutants with sulfhydryl reagents containing aromatic rings or alkyl chains should encourage hydrophobic interactions or tether the residue to the membrane. If the modified residue is normally within a hydrophobic environment when in a specific state (closed, transition or open), then that state would be stabilized. We have developed an in vivo screen that allows us to rapidly assess functional changes in the MscL channel simply by monitoring bacterial growth; channels in which the open, closed or even various transition states have been stabilized are identified. Once manipulations that lead to functional consequences are selected, the exact functional changes of MscL will be assessed by electrophysiological assays. Preliminary experiments have demonstrated the feasibility of this approach. When completed, this study will yield critical insight into how mechanosensitive channels sense and respond to membrane tension. Progress: Mechanosensitive (MS) ion channels that directly transduce mechanical stimuli are thought to underlie diverse physiological functions such as the senses of touch, hearing, proprioception, gravity sensing, and osmoregulation. Although the molecular entities of most eukaryotic MS channels still remain elusive, their bacterial counterparts have been extensively studied and thus serve today as paradigms for MS channels. Hence, we use these channels to study the mechanisms of how mechanosensors sense and interact with their lipid environment. This study has allowed us to determine some of the dynamics of these interactions upon channel stimulation and opening. The mechanosensitive channels of large conductance, MscL, and of smaller conductance, MscS, are the two major MS channels in bacteria. They effect a coordinated response to hypo-osmotic shock, serving as emergency release valves for the rapid flux of fluid and solutes. MscL from E. coli was the first to be identified, and is perhaps the best studied mechanosensor from any species. A crystal structure of the M. tuberculosis MscL has been solved, revealing a pentameric structure in a nearly-closed pore of about 4 Å. An extremely large conformational change of this protein must occur to achieve the open pore size of 30 Å estimated by conductance (the amount of ions that pass through the pore in a unit of time) and by molecular sieving experiments. Several models of the gating of MscL have been proposed to explain the drastic structural rearrangements to transition from the closed to open states. Although the models differ in details, all agree that a thinning of the channel in the plane of the membrane, accompanied by the tilting of the transmembrane domains, are necessary to allow the pentamer to achieve an open pore structure. Both of these arrangements require dynamic interactions between regions of the protein and lipids. Since X-ray crystallography of integral membrane proteins typically provide little information on their interactions with the surrounding lipids, other techniques have been used to address this issue for MscL. Utilizing some of these approaches, the second transmembrane span (TM2) of E. coli MscL, which faces the lipid membrane, has been defined. Five amino acids after the predicted end of TM2 there is a cluster of charged residues (RKKEE), which appear to be important for channel function; one E. coli MscL study showed that a C-terminal deletion at this point leads to non-functional channels, while another suggested that it serves as a pH sensor by changing the conformation of the region, and it was also suggested to interact with anionic phospholipids. Interestingly, analogous structures with a series

of charged amino acids distal to the cytoplasmic transmembrane helix has been noted in other MS channel families. Hence, this region, just cytoplasmic of TM2, is a prime candidate to be involved in the dynamic lipid-protein interactions that occur during MscL gating.

The project was basically to utilize an in vivo screen approach to determine areas that may be dynamic within the membrane. Basically, we utilized a library of individual cysteine mutations throughout the protein. Cysteine is unique in that it is reactive to sulfhydryl reagents. Thus, you can post-translationally modify any residue. Here we used modifiers that made the region more hydrophobic, i.e. wanting to go into the membrane. The assumption is that certain regions will normally go into the membrane upon activation of sensor; by forcing this region into the membrane, we may artificially cause the channel to be activated. We know from several other experiments that if this occurs in a living cell, it causes the cell to be less viable or die. Therefore, we can use this as a screen. The modifying reagents we used are: DMTS, which adds a fatty acid chain and would be very aggressive in burying the region within the membrane; MTSBn, which adds what equates to a phenylalanine to the site; and 4-HB-MTS, which is the same as previous but contains a polar hydroxyl group, thus adding what equates to a tyrosine.

In the prior years, we screened the available cysteine mutants, and generated additional cysteine mutants. In the last year, we finished the screening of the newly generated cysteine mutants, evaluated the data from the entire screen, and performed secondary screens on the “hits” that allow us to determine if the modifications lead to an overactive (gain-of-function; GOF) or dysfunctional (loss-of-function; LOF) sensor.

In the past year we have acknowledged NASA funding in three publications, have two more in press, and have a manuscript in preparation. One of the in press manuscripts is a minireview in the Biophysical Journal entitled “Sensing and responding to membrane tension: the bacterial MscL channel as a model system”. NASA funding is acknowledged, and some of the data from this project figures prominently within the review.

The data have provided a glimpse at how a mechanosensor dynamically interacts with the lipid membrane. Notably, the results of this project led to the finding that one residue appears to enter the lipid membrane in a piston-like manner upon sensor activation.

Bibliography Type:	Description: (Last Updated: 04/09/2020)
Articles in Peer-reviewed Journals	Iscla I, Wray R, Blount P. "The dynamics of protein-protein interactions between domains of MscL at the cytoplasmic-lipid interface." Channels (Austin). 2012 Jul-Aug;6(4):255-61. https://doi.org/10.4161/chan.20756 ; PubMed PMID: 22874845 ; PubMed Central PMCID: PMC3508904. (originally reported as In press, July 2012.) , Aug-2012
Articles in Peer-reviewed Journals	Iscla I, Blount P. "Sensing and responding to membrane tension: The bacterial MscL channel as a model system" Biophysical Journal. 2012 Jul 18;103(2):169-74. Review. Epub 2012 Jul 17.https://doi.org/10.1016/j.bpj.2012.06.021 ; PubMed PMID: 22853893 ; PuMed Central PMCID: PMC3400780/.(originally reported as In press, July 2012) , Jul-2012
Articles in Peer-reviewed Journals	Booth IR, Blount P. "Microbial Emergency Release Valves: the MscS and MscL families of mechanosensitive channels." J Bacteriol. 2012 Jun 8. [Epub ahead of print] PubMed PMID: 22685280 , Jun-2012
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Articles in Peer-reviewed Journals	Iscla I, Wray R, Blount P. "The oligomeric state of the truncated mechanosensitive channel of large conductance shows no variance in vivo." Protein Sci. 2011 Sep;20(9):1638-42. Epub 2011 Jul 19.http://dx.doi.org/10.1002/pro.686 ; PMID: 21739498 , Sep-2011
Articles in Peer-reviewed Journals	Yang LM, Zhong D, Blount P. "Chimeras reveal a single lipid-interface residue that controls MscL channel kinetics as well as mechanosensitivity." Cell Rep. 2013 Feb 21;3(2):520-7. https://doi.org/10.1016/j.celrep.2013.01.018 ; PubMed PMID: 23416054 ; PubMed Central PMCID: PMC3593973 , Feb-2013